Supplementary Material

Generalized Inverse Modeling of SIS-HOLZ Strain Profiles

Mana Norouzpour Ramtin Rakhsha

S1. Synthetic robustness check of the experimental $N = 5$ inversion

The experimental reanalysis in the main manuscript uses five surface-relaxed sine basis functions,

$$u(z) = \sum_{n=1}^{5} a_n \sin\left[n\pi\left(\frac{z}{t} + \frac{1}{2}\right)\right], \tag{1}$$

together with a fitted constant phase offset and a residual linear phase ramp. Because the generalized experimental solution contains two broad displacement maxima, an additional synthetic test was performed to examine whether the same $N = 5$ inversion procedure could artificially generate a double-maximum profile from a known single-maximum displacement profile.

The previously reported surface-relaxed bell-shaped displacement profile was used as the synthetic ground truth,

$$u_{\mathrm{true}}(z) = \frac{R}{2}\left[\cos\left(\frac{2\pi z}{t}\right) + 1\right], \tag{2}$$

with $R = 32.5\,\mathrm{pm}$ and $t = 166\,\mathrm{nm}$. The synthetic phase was calculated on the same diffraction-coordinate grid used for the experimental reanalysis. Gaussian phase noise with standard deviation

$$\sigma_\phi = 0.015 \text{ rad} \tag{3}$$

was added to the calculated phase profile. This noise level is comparable to the phase residual of the representative experimental $N = 5$ fit.

Fifty independent noisy phase realizations were generated. Each realization was fitted using the same $N = 5$ basis representation and the same nuisance parameters used in the experimental analysis. Three optimization starts were used for each realization. A recovered displacement profile was classified as a false double-maximum solution if two or more meaningful local maxima were detected using the criterion implemented in the supplied MATLAB code.

No false double-maximum recovery was observed in the 50 realizations. The median recovered peak projected displacement was 33.07 pm, with a 16th–84th percentile interval of 31.63 pm to 34.48 pm. The median phase RMSE was approximately 0.0139 rad. Thus, under the tested noise level and fitting conditions, the $N = 5$ inversion did not show a tendency to transform a single-maximum bell-shaped truth into a spurious double-maximum profile.

S2. MATLAB implementation and code availability

The MATLAB implementation associated with the main manuscript and this Supplementary Material is publicly available at:

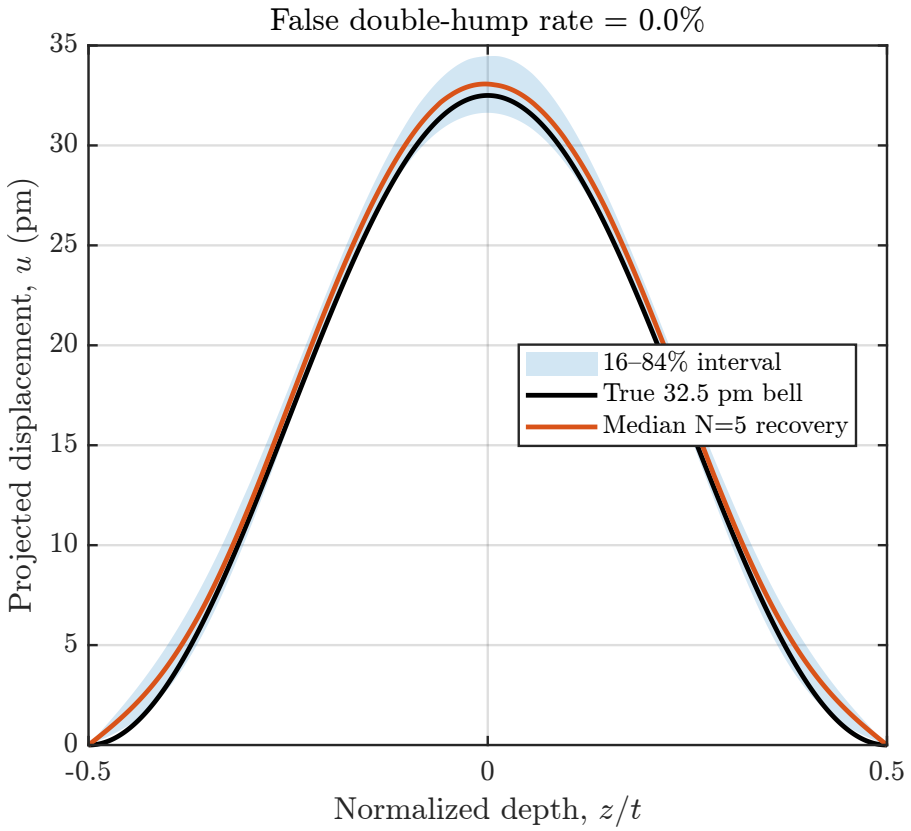


Figure 1: Synthetic robustness test of the $N = 5$ experimental inversion. The black curve is the prescribed single-maximum 32.5 pm bell-shaped displacement profile. The recovered curve represents the median of 50 independent $N = 5$ inverse solutions obtained after adding Gaussian phase noise with $\sigma_\phi = 0.015$ rad. The shaded region represents the 16th–84th percentile interval. No false double-maximum recovery was observed in the 50 realizations.

`https://github.com/RamtinRakhsha/SIS-HOLZ-Inverse-Modeling`

The repository contains the MATLAB scripts used for the forward simulations, synthetic inverse validation, robustness analyses, experimental reanalysis, model-order comparison, and the supplementary validation reported here.

The principal MATLAB files are:

- `Main_Figures/Figure1_Forward_Simulations.m`
- `Main_Figures/Figure3_Synthetic_Validation.m`
- `Main_Figures/Figure4a_Noise_Basis_Recoverability.m`
- `Main_Figures/Figure4b_Thickness_Sensitivity.m`
- `Main_Figures/Figure4c_Instrumental_Phase_Sensitivity.m`
- `Main_Figures/Figure5_Experimental_Reanalysis_N5.m`
- `Model_Order/Experimental_Model_Order_Sweep_N1_N8.m`

- `Supplementary/Supplementary_False_Double_Hump_Test_N5.m`

The original experimental phase dataset used for the experimental reanalysis is provided as:

`Data/ExPhase.txt`

S3. Reproducibility of Supplementary Fig. S1

Supplementary Fig. 1 is generated by:

`Supplementary/Supplementary_False_Double_Hump_Test_N5.m`

The script uses a fixed random-number seed so that the reported noisy realizations and summary statistics can be reproduced. The default calculation uses 50 realizations, a phase-noise standard deviation of 0.015 rad, five displacement basis functions, and three optimization starts per realization.

The supplied script writes the following outputs:

- `Fast_N5_false_double_hump_summary.csv`
- `Fast_N5_false_double_hump_profiles.pdf`
- `Fast_N5_false_double_hump_results.mat`

S4. Scope of the supplementary test

This supplementary analysis is intended as a focused robustness check. It does not establish uniqueness of the recovered experimental displacement field and does not demonstrate that the generalized experimental profile is the only profile consistent with the data. Rather, it addresses the more specific question of whether the selected $N = 5$ phase-based inversion has an intrinsic tendency to create a double-maximum displacement profile from a known single-maximum bell-shaped truth at a phase-noise level comparable to the experimental residual.

No such false double-maximum recovery was observed in the 50 realizations used in this test.